Identification of differentially expressed proteins in the prefrontal cortex of patients with Alzheimer Disease using Protein Microarrays

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INTRODUCTION

Alzheimer disease (AD) is a severe neurodegenerative disorder with high social and economic impact, which is the most common form of dementia in ageing population.

It is estimated that in 2050, approximately 80 million people worldwide will suffer from its devastating effects. Accurate and early diagnosis of AD is still difficult because early symptoms are shared with a wide variety of disorders. Definitive diagnosis requires both clinical assessment and post-mortem verification.

AD related changes along the progression of the disease selectively involves the transentorhinal cortex, spreading to the rest of the limbic system, and finally, to the more diffuse affectation of the neocortex.

HYPOTHESYS AND OBJECTIVES

A better understanding of the molecular events that take place at molecular level in AD would help identifying novel protein alterations as markers to be used in diagnosis or for treatment development.

We hypothesized that the analysis of the different brain regions of AD patients and controls might help decipher the massive proteome alterations occurring in AD, get further insights into the pathophysiology and help identifying targets and pathways for drug intervention.

In this study, we have used antibody-based quantitative proteomics for the analysis of 734 signaling molecules as a platform to increase the knowledge on the pathophysiological mechanisms altered during AD progression using tissue from the prefrontal cortex of AD patients and controls.

RESULTS

SCREENING OF HIGH-THROUGHPUT ANTIBODY MICROARRAYS

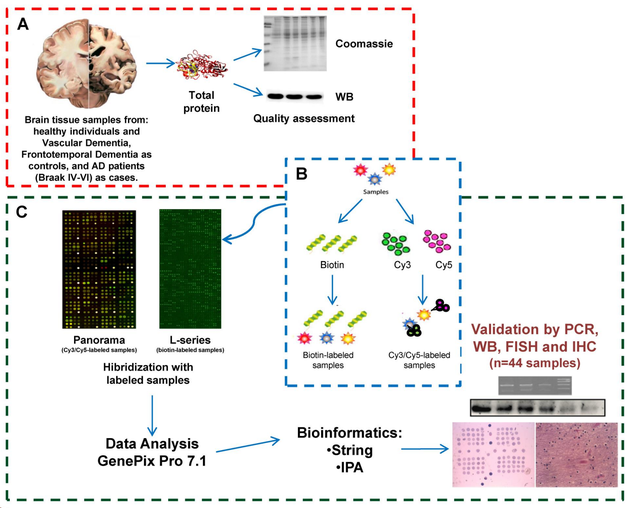


Figure 1: Schematic representation of the identification of AD-specific protein alterations non-related to other dementias by high-density antibody microarray. (A) Protein extraction. (B) Sample labeling. Pooled protein extracts of the AD Braak IV, V, and VI disease groups, and healthy individuals, FTD, and VD control groups were labeled either with Cy3/Cy5 dyes or biotin. (C) Microarray hybridization. After image quantification with GenePix Pro7.1, normalization and data analysis, those proteins significantly altered in AD in comparison to controls (healthy individuals, FTD and VD patients) were further validated using a higher number of samples by meta-analysis of large transcriptome studies, WB, and FISH and IHC with an AD-specific tissue microarray (TMA).

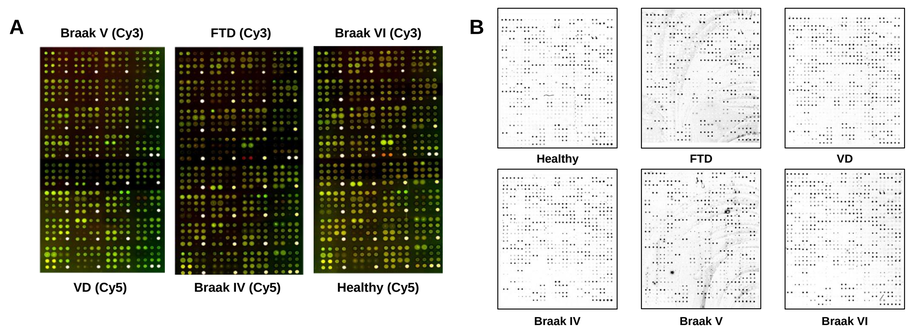


Figure 2: Representative images of the antibody microarrays used in the study to identify AD-specific protein alterations non-related to other dementias. (A) The combined images corresponding to the two fluorescence emission wavelengths of Panorama antibody microarrays incubated with equals amounts of an AD pathological group and a control group alternatively labeled with Cy3 and Cy5 to avoid any concerns related to the labeling of the samples with specific Dyes. (B) Individual performance of Label-Based Human Antibody Arrays 493 incubated separately with the indicated pooled biotin-labeled group condition followed by incubation with Cy3-streptavidin.

PATHWAY AND INTERACTOME DEREGULATED IN ALZHEIMER'S DISEASE

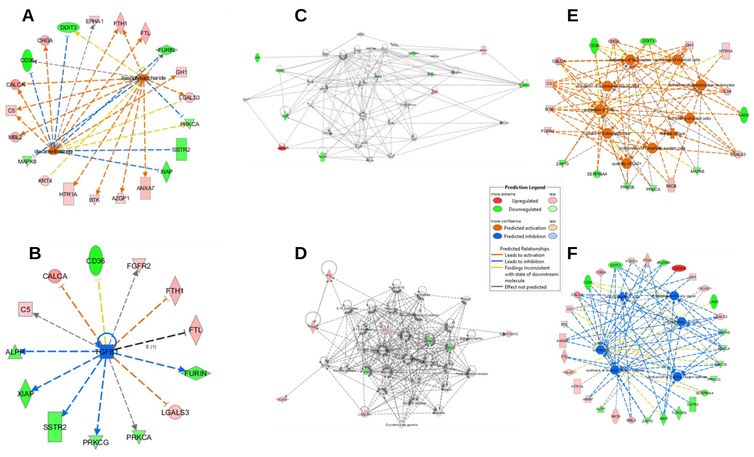


Figure 3: Ingenuity pathways analysis of the antibody microarray data. Affected protein networks were identified using the 40 differentially expressed proteins identified in the study by IPA. (A-F) Up-regulated and down-regulated proteins are shown in red and green, respectively. (A) The identified proteins were found to produce an over-activation of networks involving lipopolysaccharides (LPS) and dexamethasone. (B) A lesser activation of TGFβ-mediated networks was predicted. (C-D) Death and cellular signaling (C) and proliferation and growth (D) were the most affected networks. (E-F) Among the most affected cellular functions, cell activation and the calcium amount (E) was predicted to be activated, while production of reactive oxygen species was predicted to be inhibited (F). Fx, function; CP, cell process.

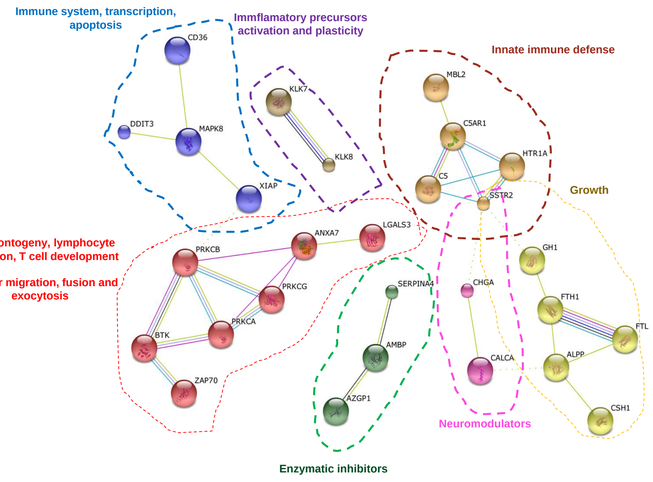


Figure 4: Protein interactome map of the differentially altered proteins in Alzheimer's Disease. STRING analysis of known and predicted protein-protein interactions of up-regulated and down-regulated gene-products differentially expressed in AD in comparison to healthy individuals and FTD and VD patients. STRING Version 9.1 and STRINGdb were used. Only those clusters of 2 or more interconnected proteins are shown. Individual nodes were removed from the figure.

VERIFICATION OF PROTEIN ALTERATIONS IN ALZHEIMER'S DISEASE

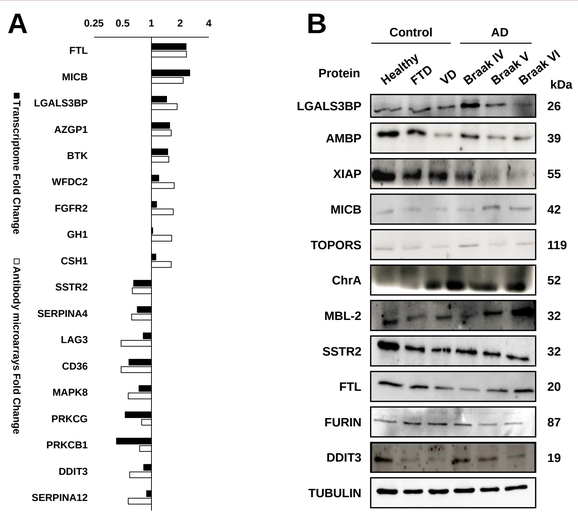


Figure 5: Validation analysis of protein alterations in tissue samples. (A) Alterations observed in a large post-mortem study examining the mRNA expression levels in prefrontal cortex tissue from late-onset AD and control patients provide data on the correlation between transcript levels and protein levels observed in 18 out of the 40 proteins identified by antibody microarray analyses. (B) A panel of antibodies against the indicated altered proteins was tested by WB analysis for verification of protein alterations using optimized dilutions of the antibodies. Tubulin was used as loading control in the same gels.

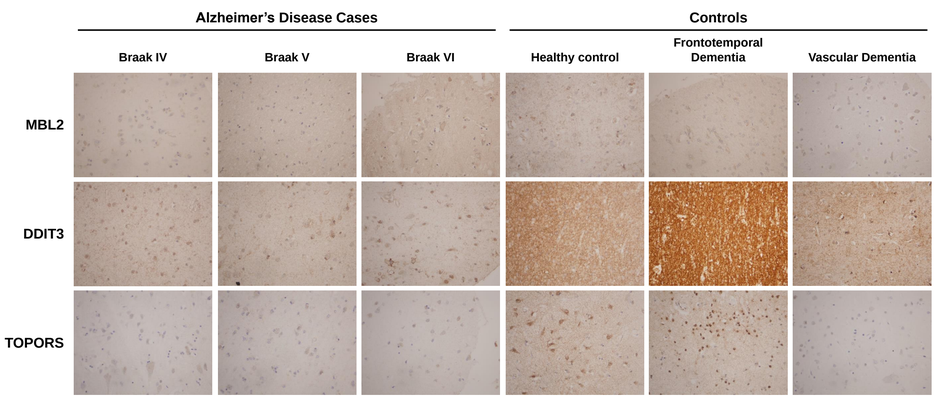


Figure 6: Immunohistochemical analysis of selected targets in AD patients and controls. Representative immunohistochemical staining pattern of indicated proteins in AD Braak stages IV, V and VI and FTD and VD patients and healthy individuals. Depicted images shown at 200× magnification images correspond to the immunohistochemical staining of TOPORS, DDIT3, and MBL2 using an AD-specific TMA containing 44 samples from AD patients and controls. These markers were selected on the basis of a significant discriminatory capacity between AD and control tissues.

CONCLUSIONS

- By comparing AD pathological groups (Braak IV, V and VI) with healthy subjects, FTD and VD patients as controls, we have identified AD-specific alterations not common to other dementias.
- Forty proteins were observed to be deregulated in Alzheimer's Disease in comparison to controls, with 63.5% of them showing up-regulation.
- Fifteen out of 40 deregulated proteins were related to the immune system, highlighting the important role of the inflammatory response in the AD pathogenesis.
- Thirteen proteins were validated by WB and immunohistochemical staining.
- Altered proteins not previously associated to AD and secreted signaling molecules constitute interesting targets for further functional analysis to determine their specific role in AD and as potential therapeutic targets of intervention and to identify blood- or CSF-biomarkers in AD.

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